

AXISYMMETRIC FLOWS WITH SWIRL OF THE H^s EULER–ARNOLD EQUATIONS: A REDUCTION, TWO THRESHOLDS, AND A MISSING WALL

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ABSTRACT. Question 7.1 of Paper I asks whether some isotropic inertia operator makes the Euler–Arnold equation on $\text{SDiff}(\mathbb{R}^3)$ or $\text{SDiff}(\mathbb{T}^3)$ blow up from smooth data. The known and conjectured blowups of the Euler equation are axisymmetric with swirl, so we reduce the family with inertia symbol $|k|^{2s}$ to that class on $\mathbb{R}^3$. The swirl circulation of the momentum is transported, and the azimuthal vorticity of the momentum is forced by one Jacobian with the angular velocity. The Lagrangian form of this forcing is the azimuthal component of Cauchy’s formula. Unlike the two-and-a-half-dimensional class of the preceding paper, the forcing keeps a Boussinesq-type linear part for every $s > 0$.

The conserved sup norm of the swirl circulation yields two scaling thresholds. It is critical at $s = 1$ for concentration on the axis, and at $s = \frac{1}{2}$ for concentration away from it. For the Boussinesq forcing with swirl odd in z , which is the parity of the known blowup candidates, the off-axis threshold drops to $s = \frac{1}{4}$. At $s = 0$, the on-axis count reproduces the scaling reported for Hou’s interior blowup candidate.

On $\mathbb{R}^3$ the geometry of the proved blowups is unavailable. Reflection in z supplies a free-slip plane, but no symmetry supplies the wall on which the Chen–Hou blowup lives. The axis takes the place of the wall, and the only remaining corner is the origin, where Hou’s interior scenario sits. We validate a spectrally exact solver against that scenario as far as a uniform grid can follow it, which is about 60% of the way to Hou’s blowup time. In that window we compare orders $s > 0$. The early growth weakens monotonically with s and reverses between $s = \frac{1}{4}$ and $s = \frac{1}{2}$.

1. INTRODUCTION

Let A be the isotropic Fourier multiplier with symbol $|k|^{2s}$. The Euler–Arnold equation

$$\partial_t m + (u \cdot \nabla)m + (\nabla u)^\top m = -\nabla p, \quad m = Au, \quad \text{div } u = 0, \quad (1)$$

is the geodesic equation of the right-invariant H^s -type metric on the volume-preserving diffeomorphisms; $s = 0$ is the Euler equation. Question 7.1 of Paper I asks whether some isotropic A gives finite-time blowup from smooth data, on $\text{SDiff}(\mathbb{T}^3)$ or on $\text{SDiff}(\mathbb{R}^3)$. A positive answer needs one construction. In [11] the family was given a continuation criterion. In [12] it was reduced to the two-and-a-half-dimensional class, where a targeted search for collapse at $s = \frac{1}{4}$ found none.

That class lacks the one ingredient of the known Euler blowups. Luo and Hou found a singularity of axisymmetric Euler flow with swirl at the wall of a cylinder [9]. Chen and Hou proved it, with computer assistance [1, 2]. The engine is centrifugal: the square of the swirl acts as a Boussinesq density. This paper carries the machinery of [12] to axisymmetric flows with swirl, on $\mathbb{R}^3$, since $\mathbb{T}^3$ has no rotational symmetry.

Results.

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- (1) *The reduction* (Section 2). With $\Gamma_m = rm^\theta$, $\Gamma_u = ru^\theta$, $\Omega = u^\theta/r$ and $w = \text{curl } m$,

$$D_t \Gamma_m = 0, \quad D_t \frac{w^\theta}{r} = \frac{1}{r} J(\Gamma_m, \Omega), \quad (2)$$

where $D_t = \partial_t + u^r \partial_r + u^z \partial_z$ and $J(f, g) = \partial_r f \partial_z g - \partial_z f \partial_r g$. The velocity is recovered by fractional powers of the operator $L_1 = \partial_r^2 + r^{-1} \partial_r - r^{-2} + \partial_z^2$. The forcing splits into a two-and-a-half-dimensional part and a Boussinesq part, $J(\Gamma_m, \Gamma_u)/r^3 + 2\Gamma_u \partial_z \Gamma_m/r^4$. The second survives for every $s > 0$.

- (2) *Structure* (Section 3). Two families of Casimirs, and the Lagrangian identity $w^\theta = r(w_0^\theta/r) \circ X_t^{-1} + J(\Gamma_m, \Phi)$, where Φ is the angle each particle has swept.
- (3) *Two thresholds* (Section 4). Fixing the conserved $\|\Gamma_m\|_{L^\infty}$, concentration on the axis is supercritical exactly for $s < 1$, and concentration away from it for $s < \frac{1}{2}$. For the Boussinesq forcing with swirl odd in z , the away-from-axis window is $s < \frac{1}{4}$. At $s = 0$ the on-axis count gives the self-similar scaling Hou reports for his interior candidate.
- (4) *The missing wall* (Section 5). On $\mathbb{R}^3$ symmetry supplies a free-slip plane but no wall. The Chen–Hou geometry is therefore unavailable, and the only blowup geometry left is the origin.
- (5) *Numerics* (Section 6). A solver in which every fractional operator is an exact symbol. It is validated against Hou’s datum as far as a uniform grid resolves it, and it compares $s > 0$ in that window.

2. THE REDUCTION

Throughout, (r, φ, z) are cylindrical coordinates on $\mathbb{R}^3$, and a field is *axisymmetric* if its cylindrical components are independent of φ . We work with smooth solutions decaying at infinity and do not revisit the local theory, which on $\mathbb{R}^3$ goes as on $\mathbb{T}^3$ [11].

Lemma 2.1 (the inertia operator on axisymmetric fields). *A commutes with rotations about the z -axis and with reflections, and so preserves the axisymmetric class and its toroidal and poloidal parts. On toroidal fields, $A(f e_\theta) = ((-L_1)^s f) e_\theta$. Hence, for $u = u^\theta e_\theta + \text{curl}(\psi e_\theta)$,*

$$m^\theta = (-L_1)^s u^\theta, \quad w^\theta = (\text{curl } m)^\theta = (-L_1)^{1+s} \psi.$$

Proof. The symbol is radial and acts on each Cartesian component, so $A[(Qv) \circ Q^{-1}] = Q(Av) \circ Q^{-1}$ for every orthogonal Q . The Cartesian Laplacian of $f(r, z) e_\theta$ is $(L_1 f) e_\theta$. So $-\Delta$ restricted to toroidal fields is unitarily equivalent to $-L_1$ on $L^2(r dr dz)$, and the functional calculus gives the formula for $A = (-\Delta)^s$. A commutes with curl , and $(\text{curl } \text{curl}(\psi e_\theta))^\theta = -L_1 \psi$. $\square$

Proposition 2.2 (the reduction). *Axisymmetric data stay axisymmetric, and (1) reduces to (2) with*

$$\Gamma_u = r(-L_1)^{-s} \left(\frac{\Gamma_m}{r} \right), \quad \psi = (-L_1)^{-1-s} w^\theta, \quad u^r = -\partial_z \psi, \quad u^z = \frac{1}{r} \partial_r(r\psi).$$

Moreover

$$\frac{1}{r} J(\Gamma_m, \Omega) = \frac{J(\Gamma_m, \Gamma_u)}{r^3} + \frac{2\Gamma_u \partial_z \Gamma_m}{r^4}. \quad (3)$$

Proof. Invariance follows from Lemma 2.1, the rotation equivariance of (1), and uniqueness.

Equation (1) says $(\partial_t + \mathcal{L}_u)m^\flat = -dp$. Evaluate it on the Killing field $\partial_\varphi = r e_\theta$. Since $[u, \partial_\varphi] = 0$ and $\partial_\varphi p = 0$, this gives $(\partial_t + u \cdot \nabla)(m^\flat(\partial_\varphi)) = 0$, and $m^\flat(\partial_\varphi) = \Gamma_m$.

The vorticity $w = \text{curl } m$ obeys $\partial_t w + (u \cdot \nabla)w = (w \cdot \nabla)u$ [11, Prop. 2.2]. Its θ -component, in cylindrical coordinates, reads $D_t w^\theta - (u^r/r)w^\theta = w^r \partial_r u^\theta + w^z \partial_z u^\theta - u^\theta w^r/r$. Insert $w^r = -\partial_z m^\theta$ and $w^z = \partial_r m^\theta + m^\theta/r$, and divide by r . With $m = m^\theta$ and $v = u^\theta$ the result is

$$D_t \frac{w^\theta}{r} = \frac{1}{r} \left(J(m, v) + \frac{\partial_z(mv)}{r} \right).$$

A direct expansion gives $J(rm, v/r) = J(m, v) + \partial_z(mv)/r$, which is the second equation of (2). Formula (3) is the same expansion written in Γ_m and Γ_u . The velocity formulas are Lemma 2.1. $\square$

At $s = 0$, $\Gamma_u = \Gamma_m = \Gamma$ and the forcing is $\partial_z(\Gamma^2)/r^4$. With $u_1 = \Gamma/r^2$ and $\omega_1 = w^\theta/r$, (2) is then exactly the system of Hou and Li used in [9, 6]. A symbolic computation (`derive_axi.py`) evaluates the full three-dimensional right-hand side of (1) from exact derivatives at random points. For generic axisymmetric fields it agrees with both equations of (2), with (3), with the vector-Laplacian identity of Lemma 2.1, and with Lemma 3.2 below, to an absolute error of at most 3.6×10^{-15} on quantities of size one to five.

The difference from the two-and-a-half-dimensional class of [12] is in (3). There the forcing was a single Jacobian, $J(\Lambda^{-2s}\theta, \theta)$, which vanishes at $s = 0$. Here the first term of (3) is of that type, but the second is a Boussinesq forcing, linear in $\partial_z \Gamma_m$, with the smoothed swirl Γ_u as its coefficient. It is the centrifugal term that drives the Luo–Hou scenario, and the metric does not remove it.

3. STRUCTURE

Proposition 3.1 (conserved quantities). *Along smooth decaying solutions the following are constant:*

$$E = \langle m, u \rangle = 2\pi \iint (m^\theta u^\theta + w^\theta \psi) r dr dz, \quad \int F(\Gamma_m) dx, \quad \int \frac{w^\theta}{r} G(\Gamma_m) dx.$$

Here F is arbitrary with $F(0) = 0$, and $G = \tilde{G}'$ with $\tilde{G}(0) = 0$.

Proof. E is the energy of [11]. The measure $dx = 2\pi r dr dz$ is preserved by the poloidal flow, and Γ_m is transported, which gives the second quantity. For the third,

$$\frac{d}{dt} \int \frac{w^\theta}{r} G(\Gamma_m) dx = 2\pi \iint J(\tilde{G}(\Gamma_m), \Omega) dr dz = -2\pi \int \tilde{G}(\Gamma_m) \partial_z \Omega \Big|_{r=0} dz = 0,$$

because $J(f, g) = \partial_r(f \partial_z g) - \partial_z(f \partial_r g)$ and $\Gamma_m = 0$ on the axis. $\square$

These are the Casimirs of the Lie–Poisson structure of axisymmetric Euler flow [5]. They depend on the bracket, not on the metric, and so they are the same for every s .

Lemma 3.2 (the Lagrangian identity). *Let Φ solve $D_t \Phi = \Omega$ with $\Phi(0) = 0$. Then*

$$w^\theta(t) = r \left(\frac{w_0^\theta}{r} \right) \circ X_t^{-1} + J(\Gamma_m(t), \Phi(t)).$$

Proof. From $\text{div } u = 0$, the poloidal divergence is $\partial_r u^r + \partial_z u^z = -u^r/r$. So $D_t J(f, g) = J(D_t f, g) + J(f, D_t g) + (u^r/r)J(f, g)$, while $D_t(1/r) = -u^r/r^2$. Hence $D_t(J(\Gamma_m, \Phi)/r) = J(\Gamma_m, \Omega)/r$. By (2), $w^\theta/r - J(\Gamma_m, \Phi)/r$ is transported, and at $t = 0$ it equals w_0^θ/r . $\square$

Φ is the azimuthal angle each particle has swept, and the lemma is the φ -component of Cauchy’s formula for the frozen-in field w . It is the analogue of the identity $q = q_0 \circ X_t^{-1} + J(\Phi, \theta)$ of [12], with the potential $\Lambda^{-2s}\theta$ replaced by the angular velocity. At $s = 0$ it expresses the classical fact that differential rotation winds the poloidal vorticity.

4. SCALING: TWO THRESHOLDS

For every $\lambda, \mu > 0$, the map $u \mapsto \mu^{-1}\lambda u(x/\lambda, t/\mu)$ sends solutions of (1) to solutions, exactly as in [12]. Then $m \mapsto \mu^{-1}\lambda^{1-2s}m(x/\lambda, t/\mu)$, and dilations about points of the axis preserve the axisymmetric class.

Proposition 4.1 (on the axis). *Under this symmetry $\Gamma_m \mapsto \mu^{-1}\lambda^{2-2s}\Gamma_m(x/\lambda, t/\mu)$. Fixing the conserved $\|\Gamma_m\|_{L^\infty}$ forces $\mu = \lambda^{2-2s}$. Concentration to scale λ on the axis therefore takes time λ^{2-2s} , which tends to zero exactly when $s < 1$.*

Proof. $\Gamma_m = rm^\theta$ carries one factor of λ from r and $\mu^{-1}\lambda^{1-2s}$ from m . $\square$

So the Casimir $\|\Gamma_m\|_{L^\infty}$ is critical at $s = 1$ for on-axis concentration. At $s = 0$ the count gives $\mu = \lambda^2$, that is, $\lambda \sim (T - t)^{1/2}$. This is exactly the self-similar scaling Hou reports for his interior candidate, a length scale of order $(T - t)^{1/2}$ [6]. For $s > 0$ the same count predicts $\lambda \sim (T - t)^{1/(2-2s)}$ for any on-axis self-similar blowup that saturates the Casimir.

Away from the axis a dilation does not preserve the class, and we record a scaling count, not a theorem. Near $r = r_0 > 0$ and at scales $\lambda \ll r_0$, freeze $r = r_0$ in (3). The second term becomes a Boussinesq source $g \partial_z \Gamma_m$, with $g = 2\Gamma_u/r_0^4$ carried by the large-scale part of Γ_u . The velocity is that of a vorticity smoothed by $2s$ derivatives. Fixing $\|\Gamma_m\|_{L^\infty}$, the balance $\partial_t \omega \sim g \partial_z \Gamma_m$ against transport by $u \sim \lambda^{1+2s}\omega$ gives $\mu \sim \lambda^{(1-2s)/2}$. The first term of (3) is the Jacobian of [12], whose count is $\mu = \lambda^{1-2s}$. Both allow concentration exactly for $s < \frac{1}{2}$. At $s = 0$ the first count is the Boussinesq value $\mu = \lambda^{1/2}$.

The Boussinesq count assumes that g is of order one where the concentration happens. Whether it depends on the parity of the swirl in z . The class is invariant under $z \mapsto -z$ with w^θ odd and Γ_m either even or odd.

- If Γ_m is even about the plane on which the concentration happens, g is of order one there, and the count stands.
- If Γ_m is odd, so is Γ_u , which therefore vanishes on the plane. Across a front of Γ_m compressed to width λ against the plane, Γ_u is only of size $\lambda^{2s}\|\Gamma_m\|_{L^\infty}$, because Λ^{-2s} acts on odd functions through the kernel $|z - z'|^{2s-1} - |z + z'|^{2s-1}$. The Boussinesq source is then weaker by λ^{2s} . The balance becomes $\lambda^{-2s}\mu^{-2} \sim \lambda^{2s-1}$, so $\mu \sim \lambda^{(1-4s)/2}$, and concentration requires $s < \frac{1}{4}$.

The Jacobian count uses only the small-scale part of Γ_u , so it does not change. At $s = 0$ parity is invisible, because only Γ_m^2 enters. The data of Luo and Hou [9] and Hou's interior datum [6] both have u^θ proportional to $\sin 2\pi z$, so both are in the odd class. For them the window of the Boussinesq mechanism is $0 < s < \frac{1}{4}$. The companion paper [13] checks $\Gamma_u \sim \lambda^{2s}$ numerically. For a one-dimensional wall model with even swirl, it proves that the window $0 < s < \frac{1}{2}$ is sharp.

Remark 4.2. The two windows differ because the axis is where r itself scales. For the blowup geometry of Luo and Hou, at a wall away from the axis, the relevant window is $0 < s < \frac{1}{2}$ for even swirl, and $0 < s < \frac{1}{4}$ for the odd swirl of their data. For an interior blowup on the axis it is $0 < s < 1$.

5. THE MISSING WALL

Proposition 5.1 (a free-slip plane). *The class in which Γ_m and w^θ are odd in z is invariant, and on it $u^z = 0$ on $\{z = 0\}$.*

Proof. L_1 commutes with $z \mapsto -z$, so Γ_u and ψ are odd with Γ_m and w^θ . Each term of (3) is then odd. Also $u^z = r^{-1}\partial_r(r\psi)$ is odd, and so it vanishes on $\{z = 0\}$. $\square$

The Luo–Hou and Chen–Hou blowups live at the intersection of this plane with a solid wall $\{r = 1\}$ on which $u^r = 0$ [9, 1, 2]. On $\mathbb{R}^3$ no symmetry of (1) produces such a wall. A symmetry class makes a surface impermeable when an isometry exchanges its two sides, as a reflection does for a plane. No isometry of $\mathbb{R}^3$ exchanges the inside and the outside of a cylinder $\{r = r_0\}$. A bounded cylinder would need A redefined with boundary conditions, and that is outside Question 7.1. The one surface on which $u^r = 0$ for every axisymmetric solution is the axis itself. So the axis takes the place of the wall, and the corner of the Luo–Hou geometry becomes the origin, the intersection of the axis with the plane of Proposition 5.1.

This is where Hou’s interior candidate sits [6]. Its status at $s = 0$ is numerical: a potential singularity at the origin, with finite energy and self-similar scaling, and no proof. A recent interval-arithmetic analysis of an interior transition [14] states its unresolved steps and claims no whole-space singularity. Section 4 places this geometry in the wider of the two windows, $0 < s < 1$. A positive answer to Question 7.1 along this route would therefore go through a mechanism that is not established even for the Euler equation. Among neighbouring results, axisymmetric Euler– α flow is understood without swirl [8]. The proved blowups of this class all use a wall [1, 2, 3].

6. NUMERICS

Method. Fields are expanded in a Fourier series in z and a Fourier–Bessel series in r , $f = \sum_n c_n J_1(k_n r)$ with $k_n = j_{1,n}/R$. Collocation is at the quasi-discrete Hankel points $r_m = j_{1,m}R/j_{1,N+1}$ [4]. In this basis $-L_1$ is diagonal, with eigenvalue $k_n^2 + k_z^2$, so $(-L_1)^{-s}$ and $(-L_1)^{-1-s}$ are exact symbols. Radial derivatives use the identities $r^{-1}\partial_r(rJ_1(kr)) = kJ_0(kr)$ and $\partial_r J_1(kr) = kJ_0(kr) - J_1(kr)/r$. We evolve Γ_m and w^θ/r with the classical fourth-order Runge–Kutta scheme and the high-order filter of [7]. A run stops by a fixed rule, when the k^2 -weighted spectral tail of m^θ or w^θ in the top 20% of either index exceeds 10^{-4} (`axisolve.py`).

On smooth axisymmetric fields vanishing at the wall, every operator is exact to 10^{-11} – 10^{-16} (`axitest.py`). Two apparent failures during testing came from test functions that were not admissible: a swirl with an r^2 component, which is a cone singularity on the axis, and a profile that did not vanish at $r = R$.

Validation against Hou’s datum at $s = 0$. Hou’s datum is $u_1 = 12000(1-r^2)^{18} \sin(2\pi z)/(1+12.5\sin^2 \pi z)$, $\omega_1 = 0$, on the cylinder $r \leq 1$ with period 1 in z [6]. We registered three criteria before the runs: conservation, agreement between resolutions, and entry into Hou’s regime $\|\omega\|_{L^\infty} \sim (T-t)^{-1}$ with T in $[0.0022, 0.0024]$. The last value Hou reports reaching is $t = 0.0022765$.

	256 ²	512 ²
resolved to	$t = 0.00130$	$t = 0.00146$
drift of E , $\int \Gamma_m^2$	2×10^{-9} , 2×10^{-10}	9×10^{-11} , 9×10^{-12}
growth of $\ \omega\ _{L^\infty}$	$\times 2.0$	$\times 2.3$

- *Conservation* holds.
- *Agreement between resolutions.* $\|\omega\|_{L^\infty}$ agrees to 1.8×10^{-4} . The axis value of u_1 , computed from the series without reference to the grid, agrees to four digits except at the end of the 256² window, where it differs by 1.5×10^{-3} .
- *Entry into Hou’s regime is not reached.* A uniform grid resolves only the first 60% of Hou’s interval, where the growth is still pre-asymptotic. Hou follows the solution with adaptive meshes. The direction is Hou’s: the maximum of $|\omega|$ sits on the plane $z = 0$ and moves toward the axis, from $r = 0.164$ to $r = 0.121$.

What the solver can be trusted with is the resolved window. We therefore use it for comparisons within that window only.

Orders $s > 0$ in the resolved window. We ran $s = 0.1, 0.25, 0.5$ and 0.75 at 512^2 from the same *velocity* datum, so that $m^\theta = (-L_1)^s u^\theta$, with the same stopping rule (`axirun.py`). Every run stopped by resolution between $t = 0.0013$ and $t = 0.0018$. That includes the larger s , whose momentum carries more fine structure than its velocity. The registered hypotheses concerned the late regime: for $s \leq \frac{1}{4}$, entry into $\|\omega\|_{L^\infty} \sim (T - t)^{-1}$; for $s \geq \frac{1}{2}$, growth of $\|\omega\|_{L^\infty}$ by less than a factor of 10 up to $t = 0.007$. Neither regime is reached, so neither hypothesis is tested.

Within the common window the comparison is clean. The table is at $t = 0.0013$. Here $u_1(0)$ is the maximum over z of the axis value, and both $\|\omega\|$ and $u_1(0)$ are relative to their initial values.

s	$\ \omega\ _{L^\infty}$	r at the maximum of $ \omega $	$u_1(0)$
0	2.01	0.135	0.672
0.1	1.90	0.135	0.637
0.25	1.55	0.143	0.531
0.5	0.89	0.244	0.416
0.75	0.53	0.345	0.334

Energy and $\int \Gamma_m^2$ are conserved to 10^{-7} or better in every run. At $s = 0.25$, the 256^2 and 512^2 runs agree on $\|\omega\|_{L^\infty}$ to 1.5×10^{-4} .

The early mechanism weakens monotonically with s . For $s \leq \frac{1}{4}$ the vorticity grows and its maximum moves toward the axis, as at $s = 0$, more slowly as s increases. For $s \geq \frac{1}{2}$ it does neither: in the window the vorticity decays, and its maximum moves away from the axis. The reversal lies between $s = \frac{1}{4}$ and $s = \frac{1}{2}$. The early growth takes place off the axis, on the plane $z = 0$ at $r \approx 0.13$ – 0.16 , where the odd swirl of this datum vanishes. The reversal therefore lies between the two off-axis counts of Section 4: $\frac{1}{4}$ for the Boussinesq forcing in the odd class, and $\frac{1}{2}$ for the Jacobian. It is far from the on-axis threshold 1. All of this concerns the first 60% of Hou’s interval, for one datum. It is an observation about the approach, not a statement about blowup.

7. DISCUSSION

The reduction shows why this class is the right one for Question 7.1: the centrifugal forcing that drives the known Euler blowups survives the metric for every $s > 0$. Scaling leaves room for concentration below $s = \frac{1}{2}$ away from the axis and below $s = 1$ on it. For the Boussinesq forcing with odd swirl, which is the parity of every datum considered here, the off-axis window shrinks to $s < \frac{1}{4}$. The wall on which the proved blowups live is not available on $\mathbb{R}^3$. What remains is Hou’s interior geometry at the origin, which is unproven even at $s = 0$.

A positive answer to Question 7.1 by this route would need, first, an interior blowup mechanism for some $s \in (0, 1)$. By Proposition 4.1, a self-similar one would concentrate like $(T - t)^{1/(2-2s)}$. Following such a mechanism numerically requires the adaptive methods of [6]; a uniform grid does not reach the asymptotic regime even at $s = 0$. We have not pursued this. What a uniform grid can contribute is the comparison of orders in Section 6. On Hou’s datum, the approach weakens monotonically with s and reverses between $s = \frac{1}{4}$ and $s = \frac{1}{2}$. If Hou’s candidate is a genuine singularity, its continuation to $s > 0$ is best tested at small s , where the early dynamics is still Hou’s. For $s \geq \frac{1}{2}$ this datum does not even begin the approach.

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The scripts underlying the computational statements are available from the author. They are `derive_axi.py`, `axisolve.py`, `axitest.py`, `axirun.py`, `axian.py` and `axianB.py`.

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